

CSCI E-106:Assignment 1

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```
library(faraway)
data(prostate, package="faraway")

summary(prostate)
```

```
##      lcavol      lweight      age      lbph
## Min.   :-1.3471  Min.    :2.375  Min.    :41.00  Min.    :-1.3863
## 1st Qu.: 0.5128  1st Qu.:3.376  1st Qu.:60.00  1st Qu.: -1.3863
## Median : 1.4469  Median :3.623  Median :65.00  Median : 0.3001
## Mean   : 1.3500  Mean   :3.653  Mean   :63.87  Mean   : 0.1004
## 3rd Qu.: 2.1270  3rd Qu.:3.878  3rd Qu.:68.00  3rd Qu.: 1.5581
## Max.   : 3.8210  Max.   :6.108  Max.   :79.00  Max.   : 2.3263
##      svi      lcp      gleason      pgg45
## Min.   :0.0000  Min.   :-1.3863  Min.    :6.000  Min.    : 0.00
## 1st Qu.:0.0000  1st Qu.: -1.3863  1st Qu.:6.000  1st Qu.: 0.00
## Median :0.0000  Median :-0.7985  Median :7.000  Median : 15.00
## Mean   :0.2165  Mean   :-0.1794  Mean   :6.753  Mean   : 24.38
## 3rd Qu.:0.0000  3rd Qu.: 1.1786  3rd Qu.:7.000  3rd Qu.: 40.00
## Max.   :1.0000  Max.    : 2.9042  Max.    :9.000  Max.    :100.00
##      lpsa
## Min.   :-0.4308
## 1st Qu.: 1.7317
## Median : 2.5915
## Mean   : 2.4784
## 3rd Qu.: 3.0564
## Max.   : 5.5829
```

a-) How many observations are in the dataset?

```
rows = nrow(prostate)
```

There are 97 observations in the dataset.

b-) Calculate the mean and median of each variable. Discuss whether any variables appear to have outliers, and briefly describe how you identified them.

```
variables = names(prostate)
response_variable = "lpsa"
explanatory_variables = variables[variables != response_variable]
```

```
stats = data.frame(
  Mean = sapply(prostate, mean),
  Median = sapply(prostate, median),
  StandardDeviation = sapply(prostate, sd)
)
print(stats)
```

```
##      Mean      Median StandardDeviation
```

```
## lcavol    1.3500096  1.446919          1.1786249
## lweight   3.6526887  3.623000          0.4966286
## age       63.8659794 65.000000          7.4451171
## lbph      0.1003558  0.300105          1.4508065
## svi       0.2164948  0.000000          0.4139949
## lcp       -0.1793637 -0.798510          1.3982478
## gleason   6.7525773  7.000000          0.7221341
## pgg45     24.3814433 15.000000          28.2040346
## lpsa      2.4783870  2.591520          1.1543287
```

A typical test for outliers selects points more than 1.5*IQR beyond the 1st/3rd quartiles.

```
stats$Q1 = sapply(prostate, quantile, na.rm = TRUE) ["25%",]
stats$Q3 = sapply(prostate, quantile, na.rm = TRUE) ["75%",]
stats$IQR = stats[, "Q3"] - stats[, "Q1"]

stats$LowerBound = stats[, "Q1"] - (1.5*stats[, "IQR"])
stats$UpperBound = stats[, "Q3"] + (1.5*stats[, "IQR"])

stats$Outliers = lapply(variables,
  function(n) {
    length(
      prostate[,n][prostate[,n] < stats[n, "LowerBound"]]
      | prostate[,n][prostate[,n] > stats[n, "UpperBound"]]
    ) })

print(stats)
```

```
##           Mean      Median StandardDeviation      Q1      Q3      IQR
## lcavol    1.3500096  1.446919          1.1786249  0.5128236  2.127041  1.614217
## lweight   3.6526887  3.623000          0.4966286  3.3759000  3.878500  0.502600
## age       63.8659794 65.000000          7.4451171 60.0000000 68.000000  8.000000
## lbph      0.1003558  0.300105          1.4508065 -1.3862940  1.558145  2.944439
## svi       0.2164948  0.000000          0.4139949  0.0000000  0.000000  0.000000
## lcp       -0.1793637 -0.798510          1.3982478 -1.3862900  1.178650  2.564940
## gleason   6.7525773  7.000000          0.7221341  6.0000000  7.000000  1.000000
## pgg45     24.3814433 15.000000          28.2040346  0.0000000 40.000000 40.000000
## lpsa      2.4783870  2.591520          1.1543287  1.7316600  3.056360  1.324700
##           LowerBound UpperBound Outliers
## lcavol    -1.908502    4.548366         0
## lweight   2.622000    4.632400         3
## age       48.000000   80.000000         0
## lbph      -5.802952    5.974804         0
## svi       0.000000    0.000000         0
## lcp       -5.233700    5.026060         0
## gleason   4.500000    8.500000         0
## pgg45     -60.000000 100.000000         0
## lpsa      -0.255390    5.043410         3
```

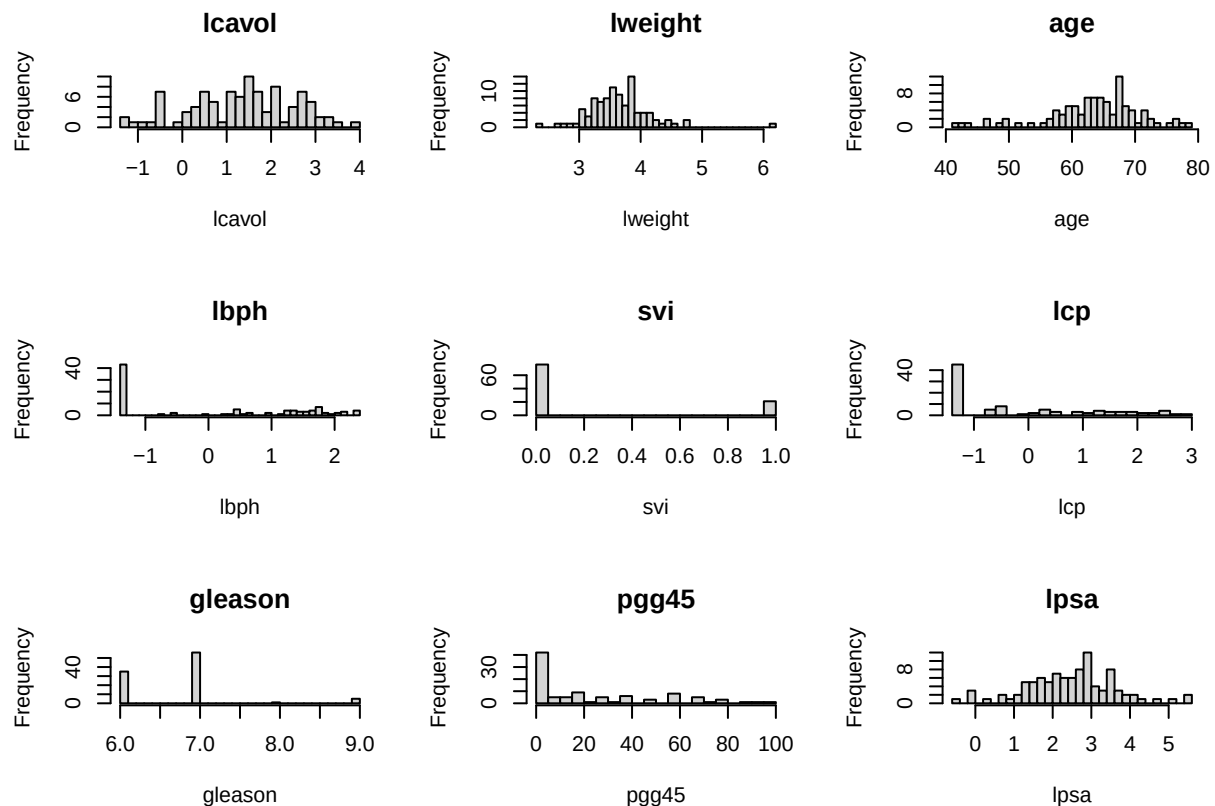
The factors `lweight` and `lpsa` have 3 outliers each.

The `pgg45` variable has a lot of 0 values, which may indicate a coding that used 0 instead of n/a.

The `gleason` variable is strongly centered around the mean of 6.753, with a standard deviation of 0.722, but about 5% of observations are 8s and 9s.

```
par(mfrow = c(ceiling(ncol(prostate)/3), 3))
for(col in variables) {
```

```
hist(prostate[[col]], main = col,
      xlab = col, breaks = 30)
}
```



c-) Compute the correlation matrix for all variables. Create scatterplots of each predictor versus the response variable lpsa. Identify which predictors appear to be strongly correlated with lpsa, using both the correlations and the plots to justify your answer.

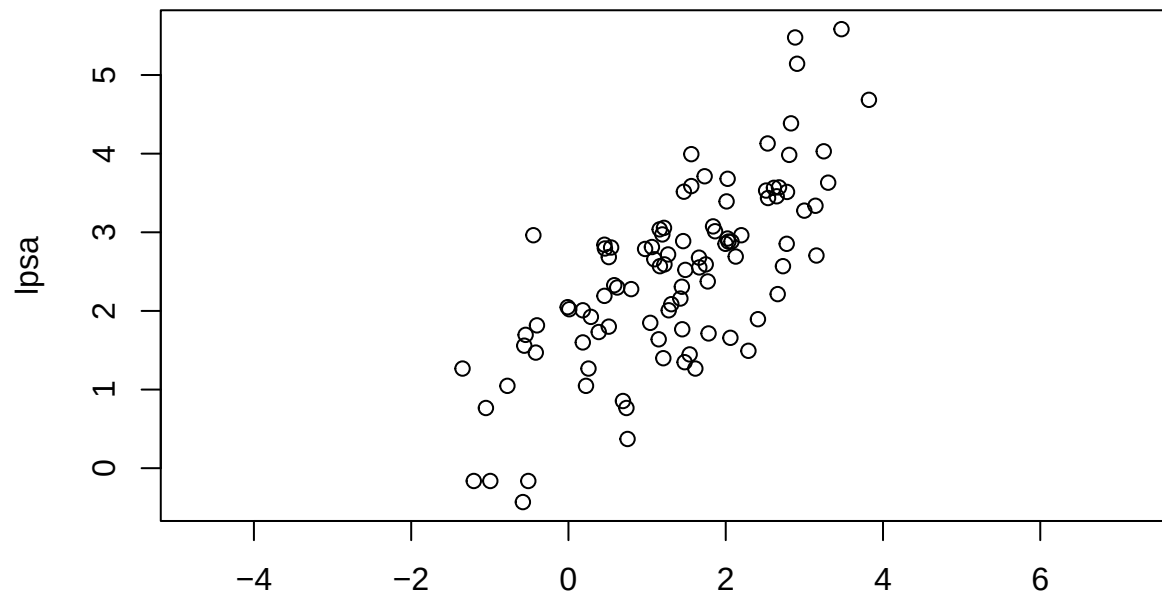
```
correlations <- cor(prostate)
print(round(correlations, 2))
```

```
##      lcavol lweight age  lbph  svi   lcp gleason pgg45 lpsa
## lcavol   1.00  0.19 0.22  0.03  0.54  0.68   0.43  0.43 0.73
## lweight  0.19  1.00 0.31  0.43  0.11  0.10   0.00  0.05 0.35
## age      0.22  0.31 1.00  0.35  0.12  0.13   0.27  0.28 0.17
## lbph     0.03  0.43 0.35  1.00 -0.09 -0.01   0.08  0.08 0.18
## svi      0.54  0.11 0.12 -0.09  1.00  0.67   0.32  0.46 0.57
## lcp      0.68  0.10 0.13 -0.01  0.67  1.00   0.51  0.63 0.55
## gleason  0.43  0.00 0.27  0.08  0.32  0.51   1.00  0.75 0.37
## pgg45    0.43  0.05 0.28  0.08  0.46  0.63   0.75  1.00 0.42
## lpsa     0.73  0.35 0.17  0.18  0.57  0.55   0.37  0.42 1.00
```

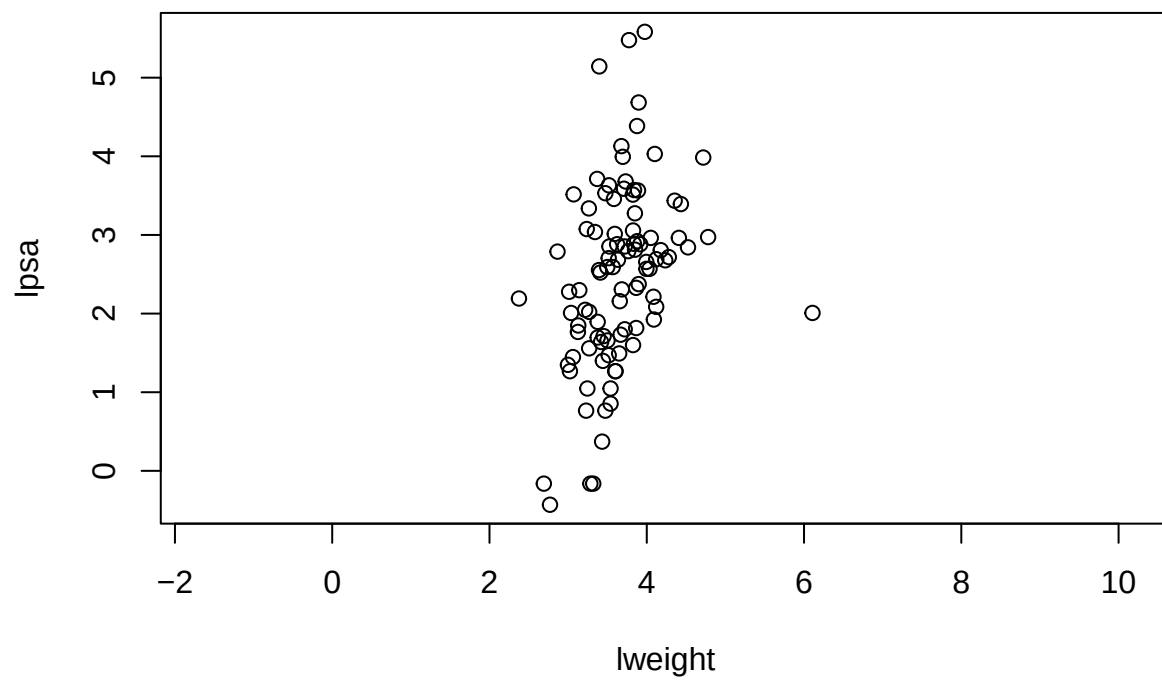
```
par(mfrow=c(1,1))
for(name in explanatory_variables) {
  plot(prostate[,name], prostate[,response_variable],
       asp=1,
       xlab = name,
       ylab = response_variable,
       main=paste("corr =", round(correlations[name, response_variable], 3)))
}
```

```
}
```

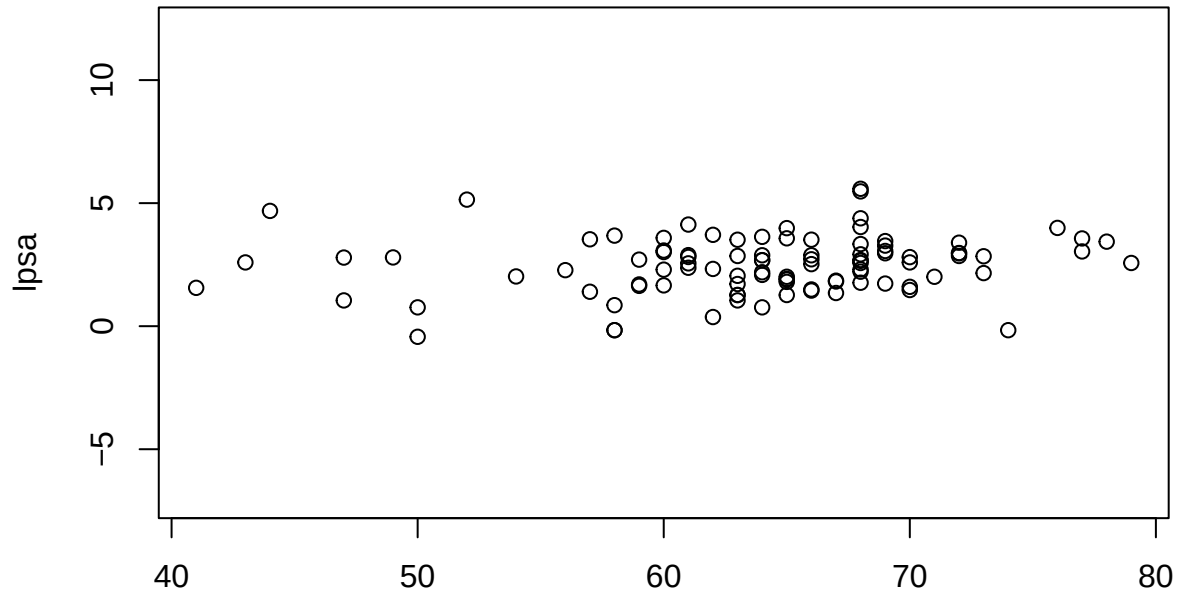
corr = 0.734



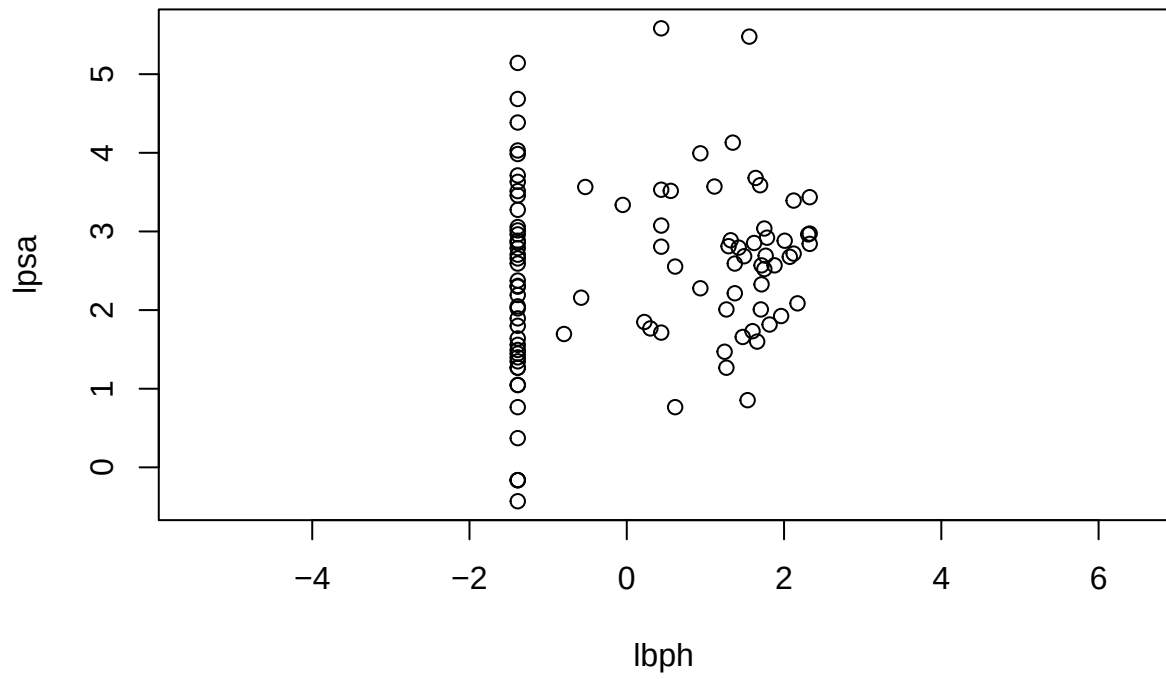
corr = 0.354



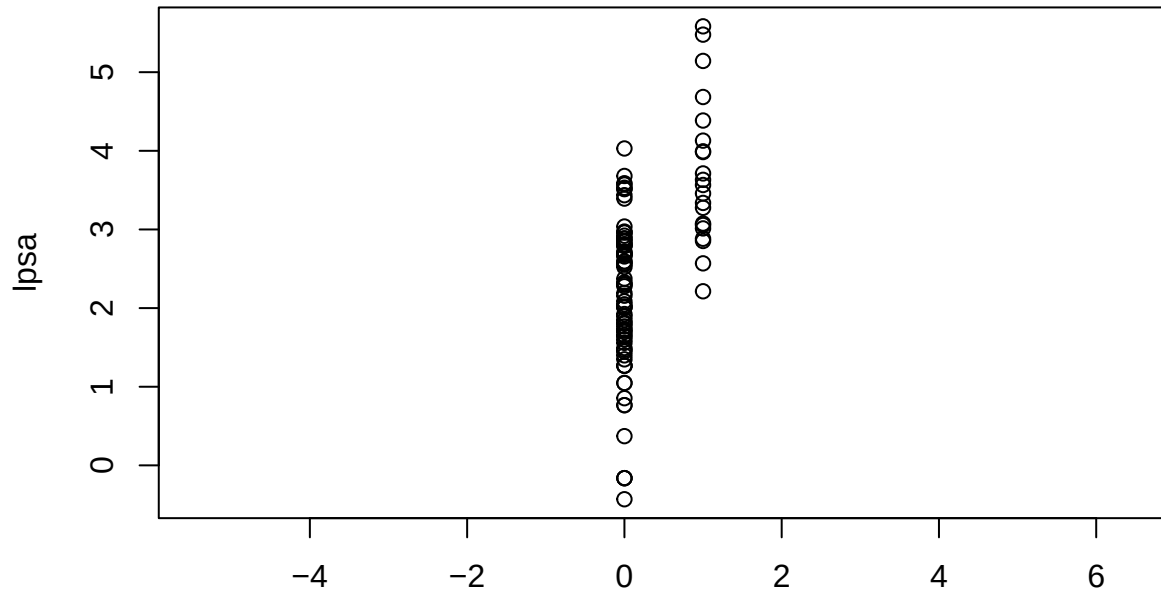
corr = 0.17



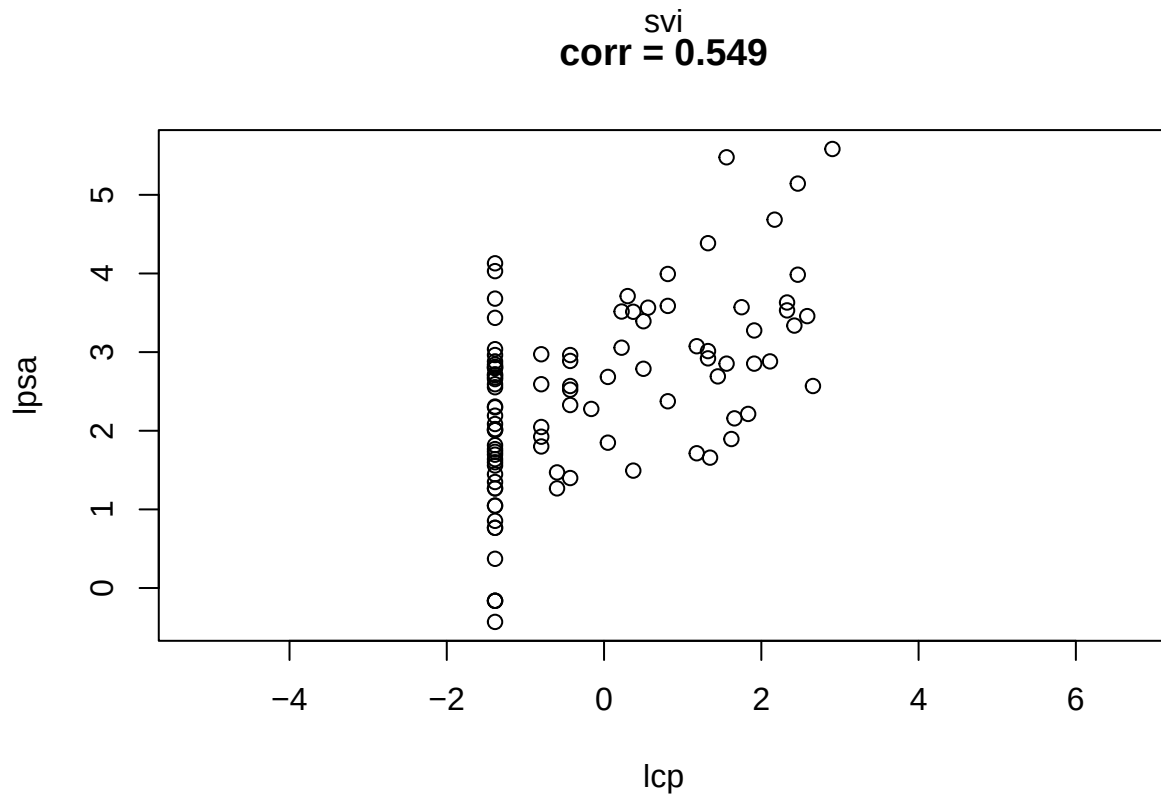
age
corr = 0.18

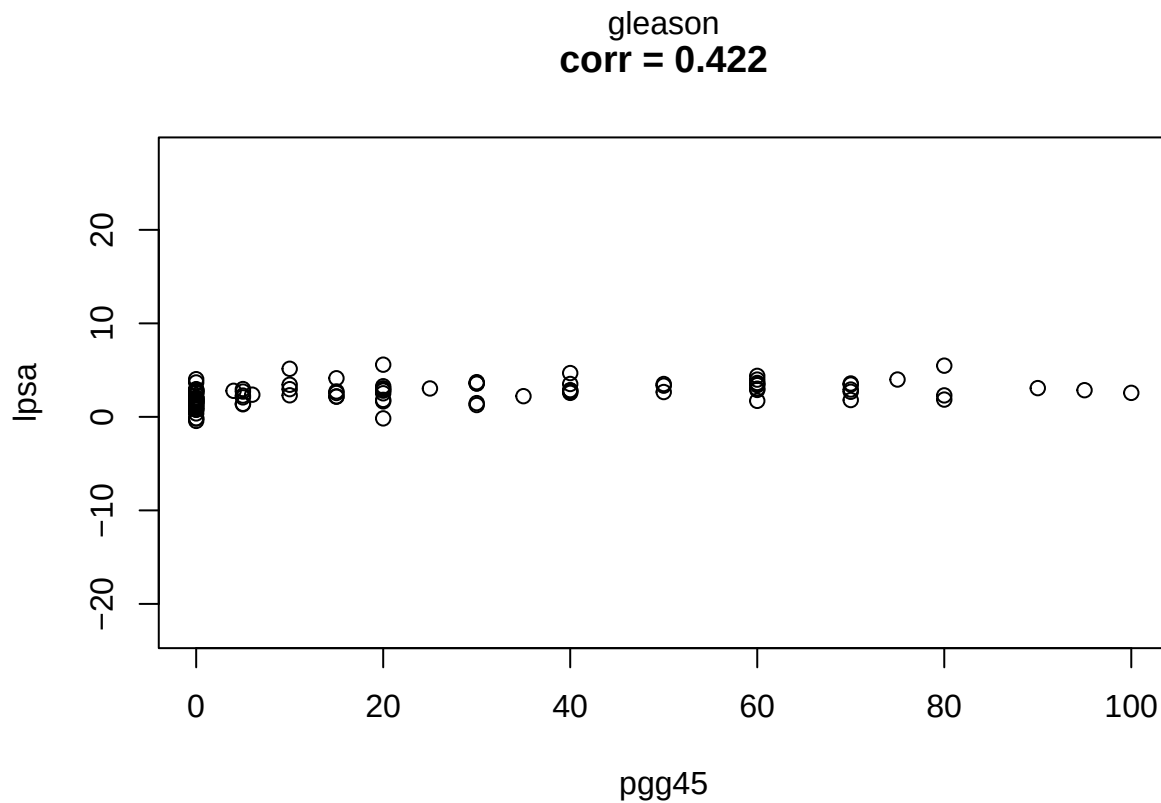
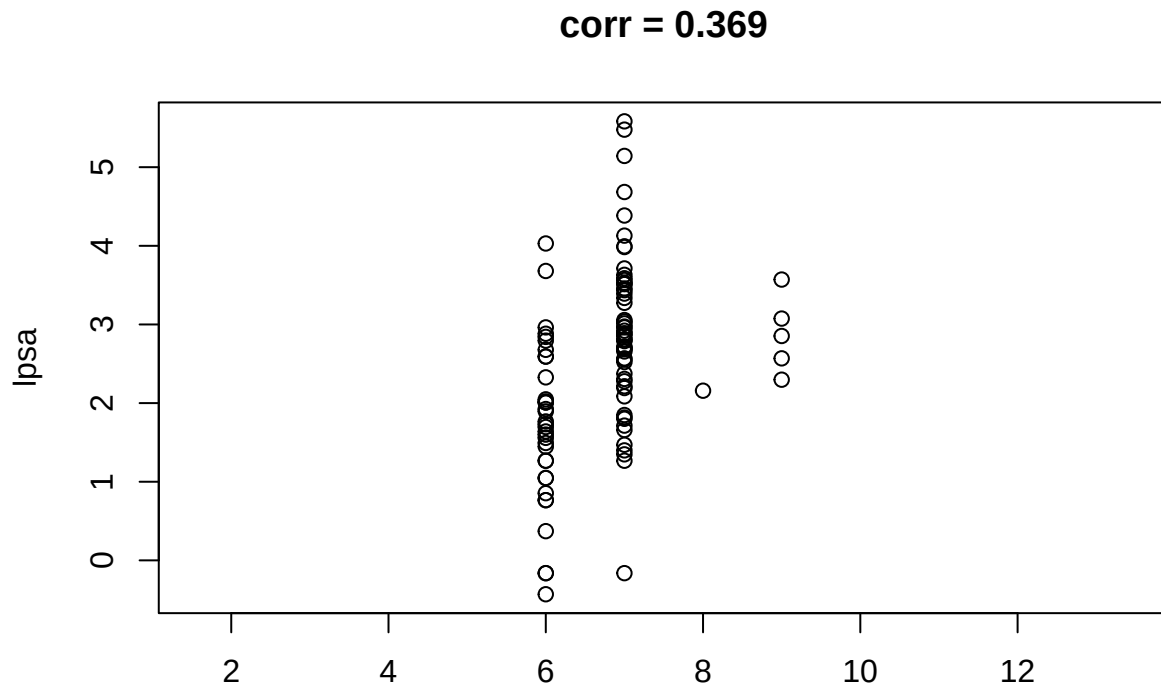


corr = 0.566



corr = 0.549





Factors **age** and **lbph** exhibit very weak to no correlation. Factors **lweight** and **gleason** exhibit a weak correlation. Factors **lci**, **lcp**, and **pgg45** exhibit a moderate correlation. Factor **lcavol** exhibits a strong correlation.

d-) Build separate simple linear regression models, by using **lcavol**, **lweight** and **age** to explain **lpsa**. For each model, clearly state the fitted regression equation with estimated coefficients.

```
lm_lccavol = lm(lpsa ~ lccavol, prostate)
summary(lm_lccavol)
```

```
##
## Call:
## lm(formula = lpsa ~ lccavol, data = prostate)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.67625 -0.41648  0.09859  0.50709  1.89673
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  1.50730     0.12194   12.36  <2e-16 ***
## lccavol       0.71932     0.06819   10.55  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.7875 on 95 degrees of freedom
## Multiple R-squared:  0.5394, Adjusted R-squared:  0.5346
## F-statistic: 111.3 on 1 and 95 DF,  p-value: < 2.2e-16
```

```
lm_lweight = lm(lpsa ~ lweight, prostate)
summary(lm_lweight)
```

```
##
## Call:
## lm(formula = lpsa ~ lweight, data = prostate)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.4908 -0.6474 -0.1022  0.6290  2.8994
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -0.5281     0.8220  -0.642 0.522120
## lweight       0.8231     0.2230   3.691 0.000373 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.085 on 95 degrees of freedom
## Multiple R-squared:  0.1254, Adjusted R-squared:  0.1162
## F-statistic: 13.62 on 1 and 95 DF,  p-value: 0.0003729
```

```
lm_age = lm(lpsa ~ age, prostate)
summary(lm_age)
```

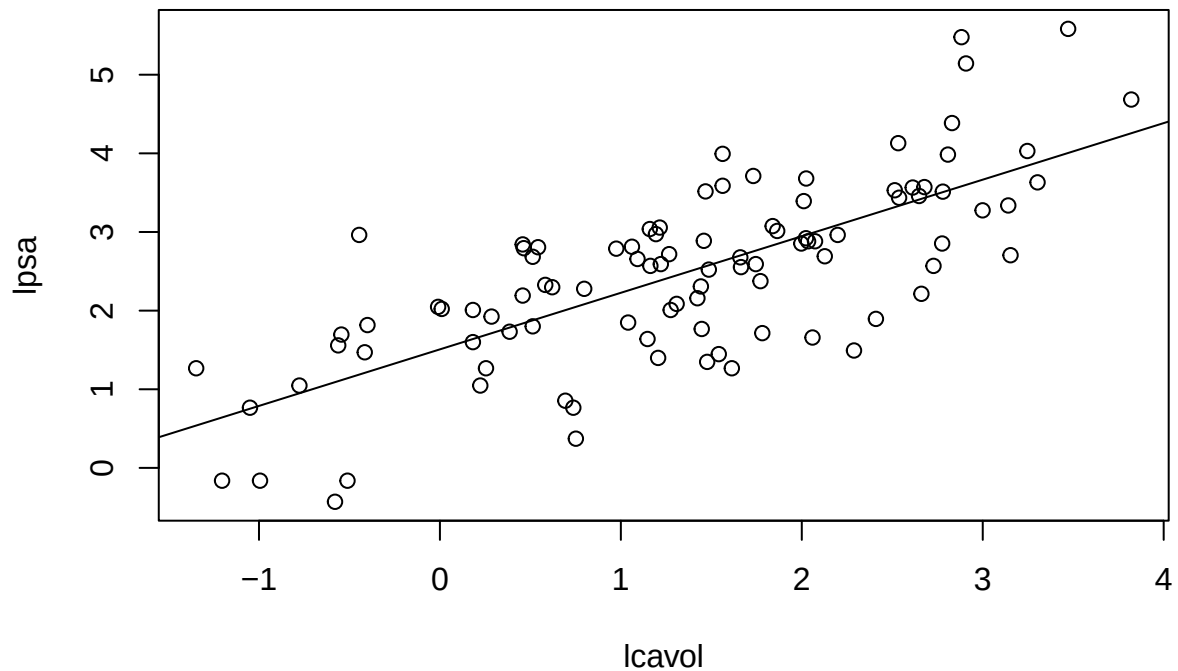
```
##
## Call:
## lm(formula = lpsa ~ age, data = prostate)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.90738 -0.71234  0.06967  0.66188  2.99584
##
```

```
## Coefficients:
##           Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.79906    1.00793   0.793  0.4299
## age          0.02629    0.01568   1.677  0.0968 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.144 on 95 degrees of freedom
## Multiple R-squared:  0.02876,    Adjusted R-squared:  0.01854
## F-statistic: 2.813 on 1 and 95 DF,  p-value: 0.09677
```

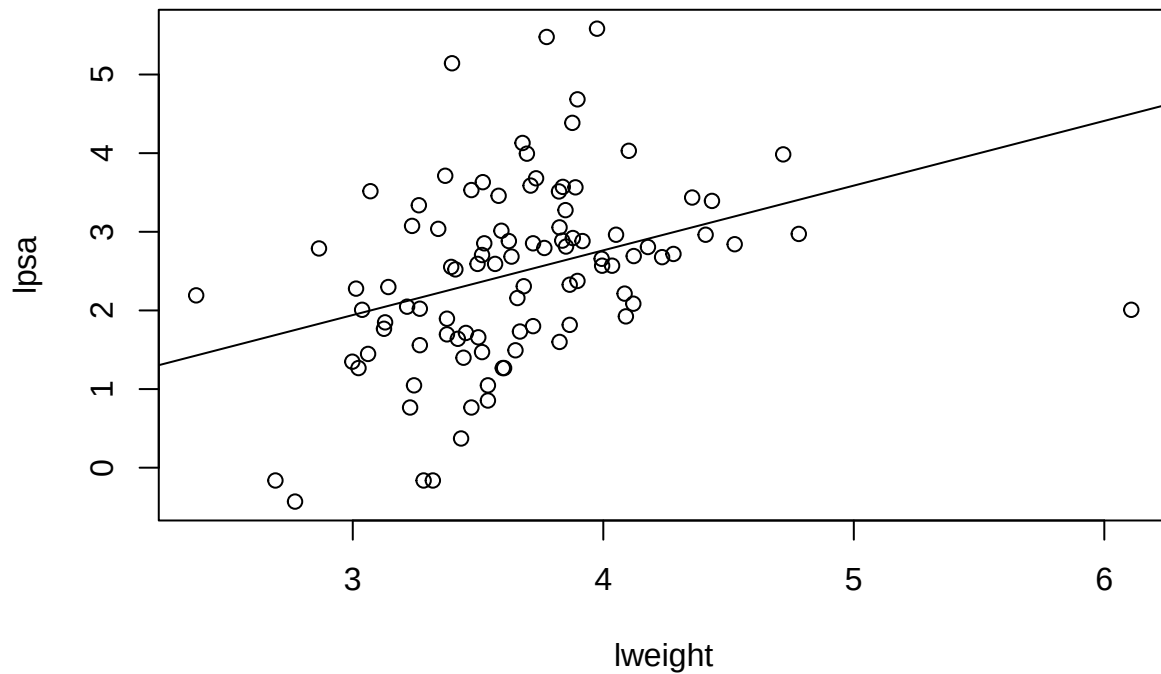
e-) For each regression model from part (d):

- Plot lpsa versus each predictor separately.
- Add the fitted regression line to each plot.
- Based on the models and plots, choose the best single-predictor model and explain your reasoning.

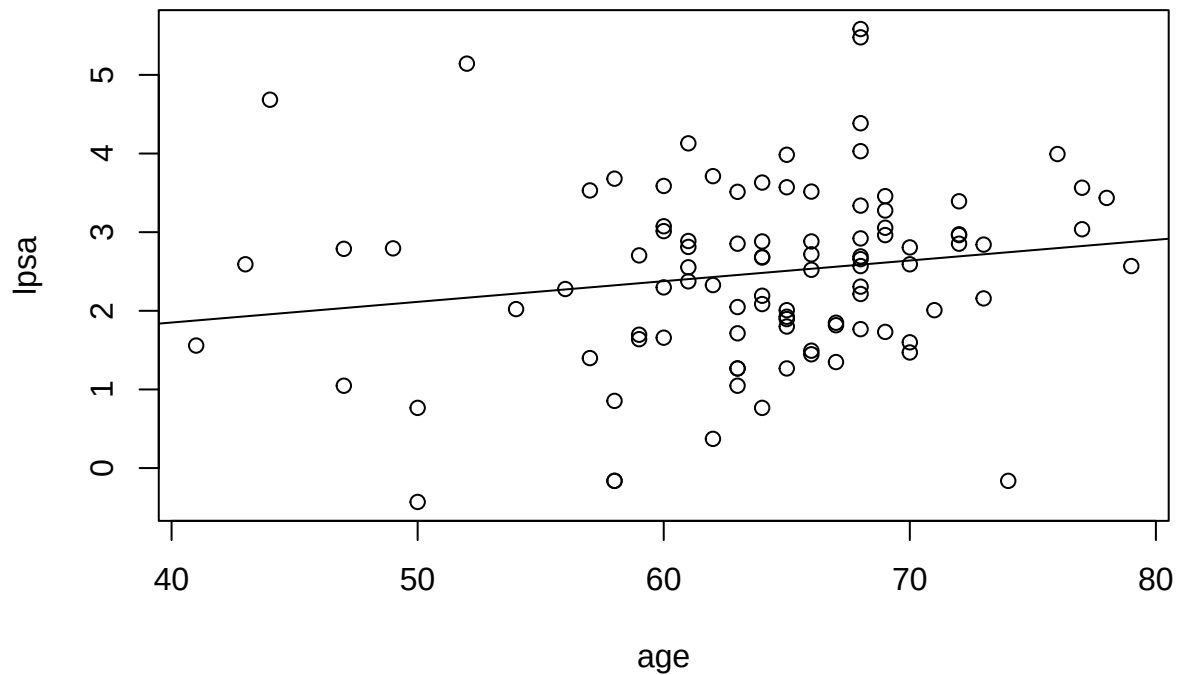
```
plot(lpsa ~ lcavol, prostate)
abline(lm_lcavol)
```



```
plot(lpsa ~ lweight, prostate)
abline(lm_lweight)
```



```
plot(lpsa ~ age, prostate)
abline(lm_age)
```



Of the three models, `lpsa ~ lcavol` has the lowest residual standard error (0.7875 vs 1.085 and 1.144).

`lpsa ~ lcavol` has the highest multiple r-squared (0.5394 vs 0.1254 and 0.02876).

`lpsa ~ lcavol` has the lowest p-value (2.2×10^{-16} vs 3.729×10^{-4} and 0.09677).

Looking at the graphs, it appears that `lpsa ~ lcavol` points fall closer to the line of best fit.

I would conclude that `lpsa ~ lcavol` is the best single-predictor model.